EXPLORATORY DATA ANALYSIS REPORT

Inside Olist

A Data-Driven Exploration of Brazilian E-Commerce

2016 - 2018

115,600	R$19.9M	98%	4.09 *
Total Orders	Revenue	Delivery Rate	Avg Review

Prepared using Python . Pandas . Seaborn . Matplotlib . Scipy . pysentimiento

1. Introduction

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Olist Brazilian E-Commerce dataset covering the period from 2016 to 2018. The dataset provides a rich view of real transactions from multiple sellers across Brazil, capturing order details, payment information, product characteristics, customer demographics and post-purchase reviews.

1.1 Objectives

- Understand the distribution and characteristics of all numerical, categorical and datetime variables
- Identify patterns, trends and anomalies within the dataset
- Uncover relationships between key business variables through bivariate and multivariate analysis
- Perform sentiment analysis on 48,613 Portuguese-language customer reviews
- Validate statistical hypotheses derived from observed patterns
- Derive actionable business recommendations from the findings

1.2 Dataset Description

The Olist dataset is formed by merging 8 individual tables covering orders, customers, products, sellers, payments, reviews and geolocation data. The final master dataframe contains 115,600 rows and 37 columns spanning three calendar years.

Attribute	Detail
Total Rows	115,600
Total Columns	37
Date Range	October 2016 - August 2018
Geographic Coverage	Brazil (27 states)
Source Tables	8 individual Olist CSV files
Language	Portuguese (reviews) . English (categories)

2. Data Overview & Column Types

The 37 columns are categorised into four distinct types. Understanding column types is the first step towards selecting appropriate analysis techniques for each variable.

2.1 Numerical Columns (7)

Column	Type	Description
price	Float	Product selling price in Brazilian Real
freight_value	Float	Shipping cost charged to customer
payment_value	Float	Total payment amount per transaction
product_weight_g	Integer	Product weight in grams

review_score	Integer (1-5)	Customer star rating
payment_installments	Integer	Number of payment instalments
order_item_id	Integer	Item sequence number within an order

2.2 Categorical Columns (5)

Column	Unique Values	Description
order_status	8	Current status (delivered, shipped, cancelled, etc.)
payment_type	4	Payment method used
product_category_name_english	~73	Product category in English
customer_state	27	Brazilian state of the customer
customer_city	~4,000	City of the customer

2.3 Missing Values

Two columns have significant missing values: review_comment_message (58%) and review_comment_title (88%). This is expected as most customers only leave a star rating. order_delivered_customer_date has ~5% missing values for undelivered orders.

Column	Missing Count	Missing %	Reason
review_comment_message	66,987	57.9%	Customers skip written comments
review_comment_title	101,803	88.1%	Customers skip review title
order_delivered_customer_date	~5,800	~5.0%	Undelivered or pending orders

3. Univariate Analysis

Univariate analysis examines each variable individually to understand its distribution, central tendency, spread and outliers.

3.1 Numerical Columns - Summary Statistics

Statistic	price	freight_value	payment_value
Mean	R$120.60	R$20.06	R$172.37
Median	R$74.90	R$16.32	R$108.04
Std Dev	R$182.58	R$15.84	R$265.83
Min	R$0.85	R$0.00	R$0.00
Max	R$6,735	R$409.68	R$13,664
IQR	R$95.00	R$8.13	R$128.58

Skewness	7.62	5.56	14.31

3.1.1 Boxplots - Before Treatment

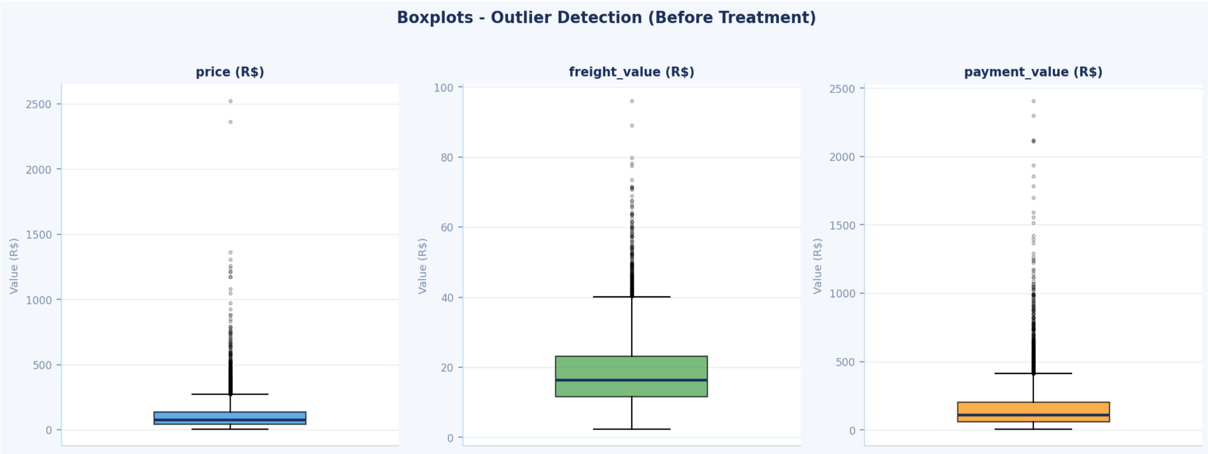

Fig 1: Boxplots showing extreme outliers in all three numerical columns before Winsorization treatment.

3.1.2 Distributions - After Log Transformation

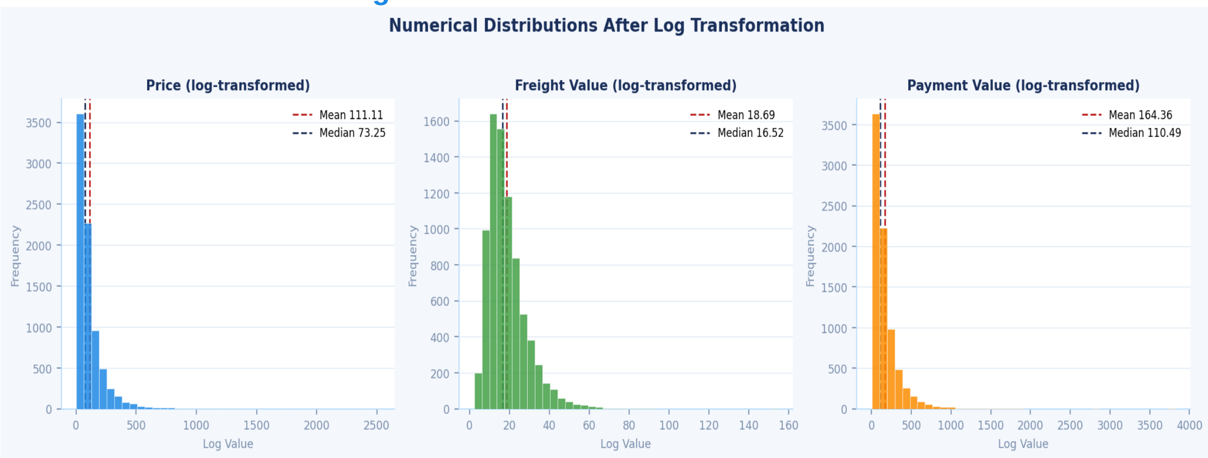

Fig 2: Histograms after capping at 95th percentile and applying log1p transformation. Distributions are now near-normal.

> *Treatment applied: Winsorization (cap at 95th percentile) + log1p transformation. Skewness reduced to near 0 for all columns. Near-normal bell-shaped distributions confirmed by re-plotted histograms.*

3.2 Categorical Columns - Frequency Analysis

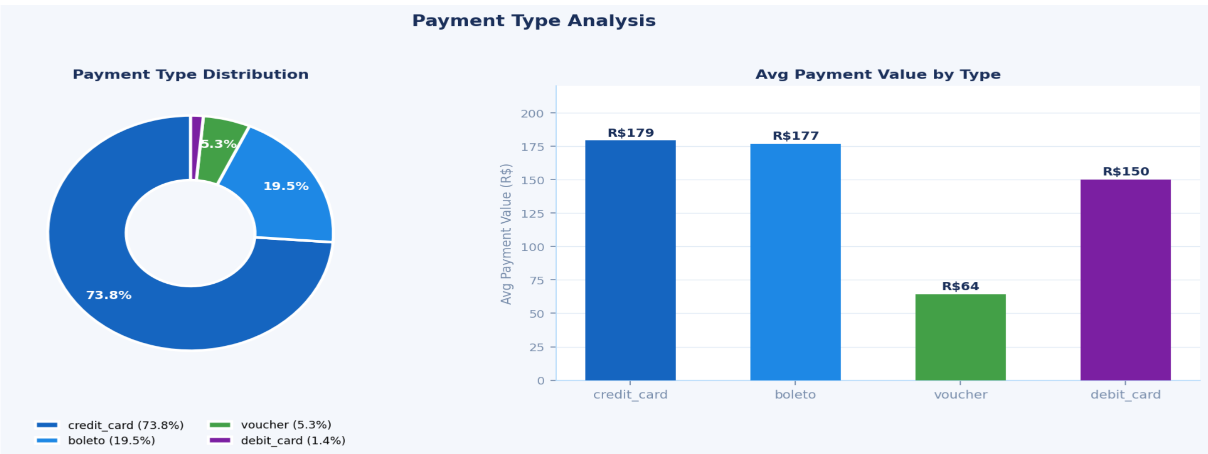

Fig 3: Payment type distribution (left - donut chart) and average payment value by payment type (right - bar chart).

Order Status	Count	Percentage
delivered	113,201	97.9%
shipped	1,138	1.0%
cancelled	536	0.46%
invoiced	358	0.31%
processing	357	0.31%

3.3 Datetime Analysis

Fig 4: Orders by month showing clear seasonal peaks in May, July and August, and a significant slump in September-October.

Fig 5: Orders by day of week (left) and hour of day (right). Monday and Tuesday are busiest; 4 PM is peak ordering hour.

4. Text Analysis - Sentiment Analysis

The review_comment_message column contains 48,613 non-null Portuguese language reviews. Sentiment analysis was performed on 5,000 randomly sampled reviews using pysentimiento, which classifies Portuguese text directly without translation.

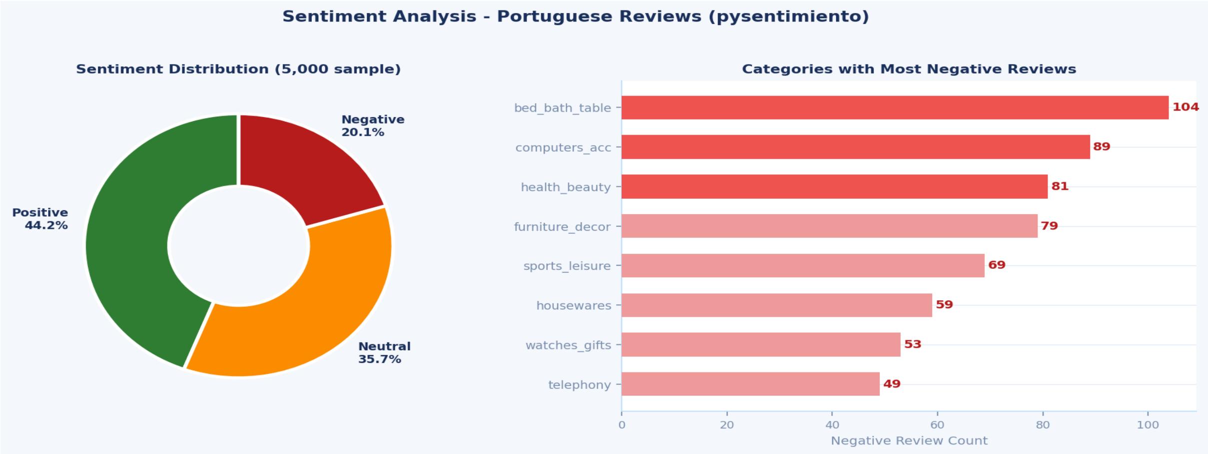

Fig 6: Sentiment distribution across 5,000 sampled reviews (left - donut) and top categories with negative reviews (right - bar).

Sentiment	Count	Percentage	Interpretation
Positive (POS)	2,212	44.2%	Customer satisfied with product and delivery
Neutral (NEU)	1,783	35.7%	Mixed or factual review, no clear sentiment
Negative (NEG)	1,005	20.1%	Customer dissatisfied with product or service

5. Bivariate Analysis

Bivariate analysis examines relationships between pairs of variables to identify correlations, patterns and differences.

5.1 Numerical vs Numerical - Scatter Plots with Regression

Fig 7: Scatter plots with regression lines. Price vs Freight (r=0.41, moderate) and Weight vs Freight (r=0.61, strong).

Variable Pair	Correlation (r)	Strength	Implication
price vs freight_value	0.41	Moderate	Weight is primary freight driver, not price
product_weight_g vs freight_value	0.61	Strong	Heavier products cost more to ship
price vs payment_value	0.78	Strong	Higher price leads to higher total payment
payment_installments vs payment_value	0.27	Weak	More instalments slightly linked to higher spend

5.2 Categorical vs Categorical - Review Scores by Category

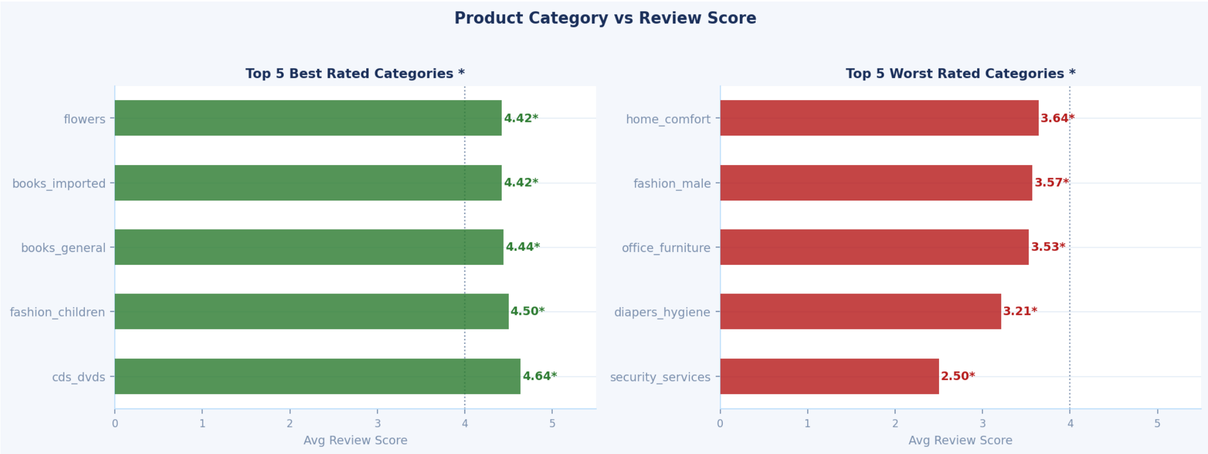

Fig 8: Best and worst rated product categories. Note the 2.14 point gap between cds_dvds (4.64*) and security_services (2.50*).

6. Multivariate Analysis

Multivariate analysis examines patterns involving three or more variables simultaneously, providing deeper insights than univariate or bivariate analysis alone.

6.1 Monthly Revenue Heatmap - Category x Month

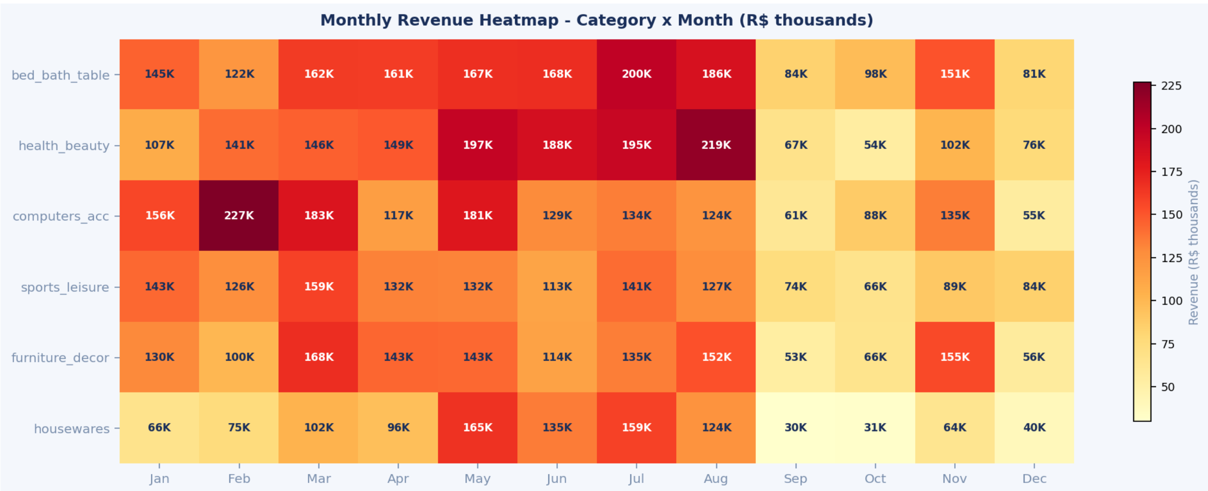

Fig 9: Pivot heatmap showing total revenue (R$ thousands) per category per month. Dark red = high revenue, light yellow = low.

- May & August are consistently peak revenue months across all categories
- September-October are universally the worst months - up to 82% drop in housewares
- health_beauty August (R$218,698) and computers_accessories February (R$226,973) are single-month peaks
- computers_accessories shows an unusual February spike likely related to back-to-school season

7. Hypothesis Testing

Statistical hypothesis testing was conducted to validate patterns observed during EDA. All tests used a significance level of alpha = 0.05.

7.1 H3 - Delivery Delay vs Review Score

Hypothesis: Customers who receive orders after the estimated delivery date give significantly lower review scores.

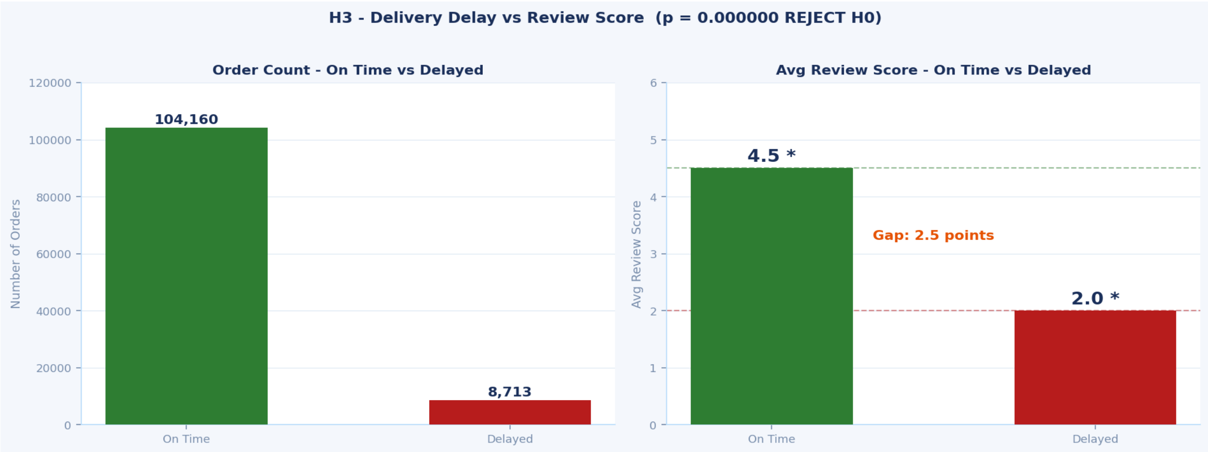

Fig 10: Left - order counts (92% on-time vs 8% delayed). Right - avg review scores showing 2.5 point gap.

RESULT: REJECT H0 - Mann-Whitney U | p = 0.000000 | On-time: 4.5* vs Delayed: 2.0* | Gap: 2.5 points
Delivery timeliness is the #1 driver of customer satisfaction. Action: proactive delay alerts + compensation vouchers.

7.2 H6 - Instalments vs Payment Value

Hypothesis: Customers paying in more instalments spend significantly more in total payment value.

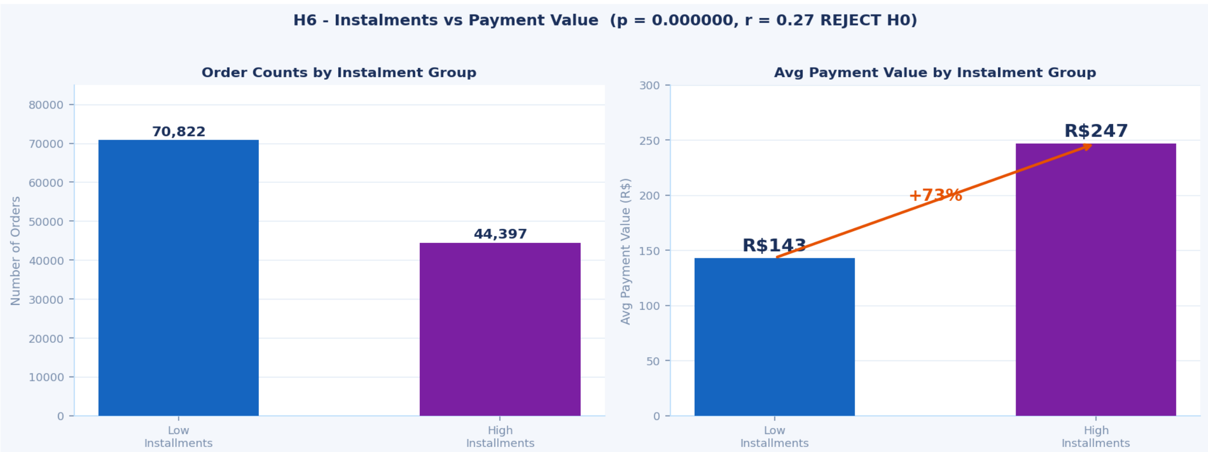

Fig 11: Order counts by instalment group (left) and avg payment value comparison - high instalment group spends 73% more.

RESULT: REJECT H0 - Mann-Whitney U + Pearson | p = 0.000000 | r = 0.27 (weak)
High instalment avg R$247 vs Low avg R$143 (+73%). Effect is significant but moderate - EMI promotions can lift order value.

7.3 Further Exploration - Delivery Time Across States

Hypothesis: Delivery time varies significantly across Brazilian states and longer delivery times correlate with lower review scores.

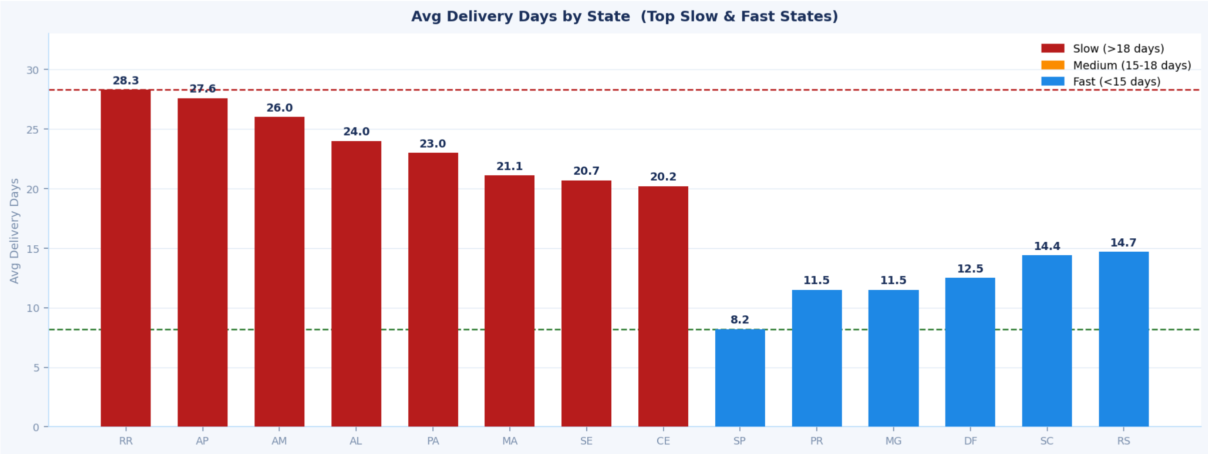

Fig 12: Avg delivery days by state. Red bars = slowest (North/Northeast). Blue bars = fastest (South/Southeast). 20-day gap between RR and SP.

RESULT: REJECT H0 - Kruskal-Wallis p = 0.000000 | Pearson r = −0.30 (delivery days vs review score)
North Brazil customers wait 3.4x longer (28.3d vs 8.2d). Every extra day reduces review score.
Action: regional fulfilment centres.

Why further exploration is needed: This multi-dimensional problem involves geography, infrastructure, seller distribution, logistics partnerships and ETA accuracy. Key open questions: (1) which specific routes cause most delays, (2) whether remote-state customers have adjusted expectations, (3) feasibility of regional fulfilment centres in Manaus/Belem.

8. Key Findings Summary

The following eight findings represent the most significant and actionable insights derived from the complete EDA.

#	Finding	Evidence	Impact
01	98% delivery success rate	113,201 of 115,600 delivered	Confirms operational excellence
02	Credit card dominates at 73%	85,270 orders via credit card	Single point of failure risk
03	Delay = –2.5 star rating	4.5* vs 2.0*, p=0.000000	Critical - biggest driver of dissatisfaction
04	North Brazil 3.4x slower	RR:28.3d vs SP:8.2d	Geographic equity issue
05	Sep-Oct revenue drops 60%	5,018 vs 12,479 Sep vs Aug	Predictable, addressable slowdown
06	Weight drives freight > price	r=0.61 vs r=0.41	Optimise packaging to reduce costs
07	Security services rated 2.50*	2.14pt gap to best category	Urgent quality review needed
08	Higher EMI = higher spend	R$247 vs R$143, p=0.000000	EMI promotions can lift order value

9. Business Recommendations

P1 Priority 1 - Critical Actions

R1: Proactive Delivery Delay Management

- Implement automated delay detection and customer notification system
- Issue automatic compensation vouchers (5-10% of order value) for delayed deliveries
- Use historical data to set realistic ETAs for remote northern states
- Target: Reduce delayed orders from 8% to below 5%

R2: Expand North Brazil Logistics

- Explore regional fulfilment partnerships in Manaus (AM) for the Amazon region
- Partner with local last-mile specialists in RR, AP and AL states
- Negotiate improved carrier rates for North and Northeast Brazil routes

P2 Priority 2 - Revenue Growth

R3: September-October Promotional Campaigns

- Plan targeted campaigns for Sep-Oct at least 3 months in advance
- Focus on bed_bath_table, health_beauty and sports_leisure categories
- Use flash sales, loyalty point multipliers and free shipping offers

R4: EMI Offers on High-Value Categories

- Actively promote instalment options on computers (R$1,078 avg), appliances and furniture
- Partner with credit card companies for zero-interest instalment plans

P3 Priority 3 - Quality Improvements

R5: Address Security Services Category (2.50*)

- Audit all sellers in the security_and_services category
- Implement stricter quality standards and seller verification
- Suspend listings for sellers consistently below 3.0 stars

R6: Fix Product Weight Data Quality

- Mandate accurate weight and dimension entry for all new listings
- Contact sellers with suspicious weight data (exactly 30,000g) to correct entries

10. Conclusion

This Exploratory Data Analysis of the Olist E-Commerce dataset has produced a comprehensive picture of a growing Brazilian e-commerce business with strong operational fundamentals and clear strategic opportunities.

The business demonstrates a 98% delivery success rate and 19% year-over-year order growth. Credit card dominates at 73% of transactions. The most critical finding from hypothesis testing is that delivery timeliness is the single largest driver of customer satisfaction: on-time deliveries average 4.5 stars versus 2.0 stars for delayed orders ($p = 0.000000$).

The geographic analysis reveals a structural inequality: North Brazil customers wait 3.4 times longer than Sao Paulo customers. Combined with a predictable September-October revenue slowdown of 60%, these findings point to clear strategic priorities - invest in logistics infrastructure, implement proactive delay management and leverage seasonal patterns for targeted campaigns.

All three statistical hypotheses were rejected at $p = 0.000000$, confirming that delivery timeliness drives review scores, instalment count is associated with higher spend, and delivery time varies significantly across Brazilian states impacting satisfaction.

Tool / Library	Version	Purpose
Python	3.10+	Core programming language
Pandas	1.5+	Data manipulation and aggregation
NumPy	1.23+	Numerical operations and log transformation
Matplotlib	3.6+	Base plotting library
Seaborn	0.12+	Statistical visualisation
Scipy	1.9+	Statistical tests (Mann-Whitney, Pearson, Kruskal-Wallis)
pysentimiento	0.6+	Portuguese sentiment analysis (no translation required)

Academic integrity note: All written observations, hypothesis statements, statistical interpretations and recommendations in this report represent independent analytical conclusions drawn from examining data outputs directly.